***Appendix 2 Search Strategy***

A.1.1 Ovid Medline search strategy

Database(s): **Ovid MEDLINE® and Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Daily and Versions(R)** 1946 to April 02, 2020
Search date: 04 April 2020

Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp Pregnancy/ | 884,967 |
| 2 | (pregnan* or maternal or gestation* or antenatal or prenatal or perinatal or natal or gravidit* or gravida* or multigravid* or primigravid*).mp. | 1,207,647 |
| 3 | (parity or parous or primiparity or multiparity or primiparous or multiparous).mp. | 60,304 |
| 4 | (expectant adj (mother* or wom#n or female*)).mp. | 1,304 |
| 5 | 1 or 2 or 3 or 4 | 1,234,652 |
| 6 | (sleep* or slept or lying down or laid down or lie down or lay down or recumben* or reclining or reclined).mp. | 212,880 |
| 7 | ((supine or prone) adj3 (position* or posture*)).mp. | 24,056 |
| 8 | ((lying or rest or rested or resting) and (position* or posture*)).mp. | 18,718 |
| 9 | 6 or 7 or 8 | 248,516 |
| 10 | 5 and 9 | 9,426 |
| 11 | Developing Countries/ | 74,062 |
| 12 | (Afghanistan* or Albania* or Algeria* or Angola* or Argentina* or Armenia* or Azerbaijan* or Bangladesh* or Belarus* or Beliz* or Benin* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Botswan* or Brazil* or Bulgaria* or Burkina* or Burundi* or Cabo Verde* or Cape Verde* or Cambodia* or Cameroon* or Central African or Chad* or China or Chinese or Colombia* or Comor* or Congo* or Costa Rica* or Cote d'Ivoir* or Ivory coast or Cuba* or Djibouti* or Dominica* or Ecuador* or Egypt* or El Salvador* or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Georgia* or Ghana* or Grenad* or Guatemala* or Guinea* or Guyan* or Haiti* or Hondura* or Hungar* or India* or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kenya* or Kiribati* or Korea* or Kosov* or Kyrgyz Republic or Lao* or Leban* or Lesotho* or Liberia* or Libya* or Macedonia* or Madagascar* or Malawi* or Malaysia* or Maldiv* or Mali* or Marshall Island* or Mauritania* or Mauriti* or Mexic* or Micronesia* or Moldova* or Mongolia* or Montenegr* or Morocc* or Mozambi* or Myanma* or Burmese or Namibia* or Nepal* or Nicaragua* or Niger* or Nigeria* or Pakistan* or Palau* or Panama* or Papua New Guinea* or Paraguay* or Peru* or Philippines or Filipino or Romania* or Rwanda* or Samoa* or | 2,646,441 |

| | | |
|---|---|---|
| | Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leon* or Soloman Island* or Somalia* or South Africa* or Sudan* or Sri Lanka* or St Lucia* or St Vincent or Grenadines or Surinam* or Swazi* or Syria* or Tajikistan* or Tanzania* or Thai* or Timor* or Togo* or Tonga* or Tunisia* or Turk* or Turkmenistan* or Tuvalu* or Uganda* or Ukrain* or Uzbekistan* or Vanuatu* or Venezuela* or Vietnam* or West Bank or Gaza or Yemen* or Zambia* or Zimbabwe*).mp. | |
| 13 | exp africa/ or exp caribbean region/ or exp central america/ or latin america/ or exp south america/ or asia/ or exp asia, central/ or exp asia, southeastern/ or exp asia, western/ or exp indian ocean islands/ or pacific islands/ or exp melanesia/ or exp micronesia/ or exp west indies/ | 845,891 |
| 14 | (africa* or asia* or caribbean or central america* or latin america* or south america* or melanesia* or micronesia* or polynesia*).mp. | 557,464 |
| 15 | (resource-limit* or resource-poor or low-resource* or limited-resource* or resource-constrain* or constrain*-resource* or under-resource* or poor*-resource* or resource-scarce* or scarce*-resource* or low-income or middle-income or lowincome or middleincome or (low adj3 middle-income)).mp. | 83,416 |
| 16 | ((developing or underdeveloped or under-developed or emerging or less-developed or least-developed or less-economically developed or least-economically developed or less-affluent or least-affluent or deprived or poor) adj (country or countries or nation or nations or region or regions or economy or economies)).mp. | 131,753 |
| 17 | ((developing or underdeveloped or under-developed or less-developed or least-developed) adj (population* or world)).mp. | 9,050 |
| 18 | (third-world* or thirdworld* or 3rd-world* or lmic or lmics or lami countr* or lalmi countr* or transitional countr*).mp. | 8,007 |
| 19 | (low* adj (gdp or gnp or gross domestic or gross national)).mp. | 239 |
| 20 | ((underserved or under served) adj (countr* or nation? or population*)).mp. | 3,057 |
| 21 | 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 | 3,084,106 |
| 22 | 10 and 21 | 1,374 |
| 23 | exp animals/ not humans.sh. | 4,685,426 |
| 24 | 22 not 23 | 1,297 |
| 25 | Exp Pregnancy, Animal/ | 25,449 |
| 26 | 24 not 25 | 1,297 |

### A.1.2 Ovid Embase search strategy

Database(s): **Embase Classic+Embase** 1947 to 2020 April 02
Search date: 04 April 2020

Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp pregnancy/ | 795,893 |
| 2 | (pregnan* or maternal or gestation* or antenatal or prenatal or perinatal or natal or gravidit* or gravida* or multigravid* or primigravid*).mp. | 1,492,153 |

| | | |
|---|---|---|
| 3 | (parity or parous or primiparity or multiparity or primiparous or multiparous).mp. | 72,777 |
| 4 | (expectant adj (mother* or wom#n or female*)).mp. | 1,894 |
| 5 | 1 or 2 or 3 or 4 | 1,518,394 |
| 6 | (sleep* or slept or lying down or laid down or lie down or lay down or recumben* or reclining or reclined).mp. | 358,311 |
| 7 | ((supine or prone) adj3 (position* or posture*)).mp. | 41,080 |
| 8 | ((lying or rest or rested or resting) and (position* or posture*)).mp. | 28,155 |
| 9 | 6 or 7 or 8 | 414,788 |
| 10 | 5 and 9 | 16,219 |
| 11 | exp developing country/ | 95,379 |
| 12 | (Afghanistan* or Albania* or Algeria* or Angola* or Argentina* or Armenia* or Azerbaijan* or Bangladesh* or Belarus* or Beliz* or Benin* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Botswan* or Brazil* or Bulgaria* or Burkina* or Burundi* or Cabo Verde* or Cape Verde* or Cambodia* or Cameroon* or Central African or Chad* or China or Chinese or Colombia* or Comor* or Congo* or Costa Rica* or Cote d'Ivoir* or Ivory coast or Cuba* or Djibouti* or Dominica* or Ecuador* or Egypt* or El Salvador* or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Georgia* or Ghana* or Grenad* or Guatemala* or Guinea* or Guyan* or Haiti* or Hondura* or Hungar* or India* or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kenya* or Kiribati* or Korea* or Kosov* or Kyrgyz Republic or Lao* or Leban* or Lesotho* or Liberia* or Libya* or Macedonia* or Madagascar* or Malawi* or Malaysia* or Maldiv* or Mali* or Marshall Island* or Mauritania* or Mauriti* or Mexic* or Micronesia* or Moldova* or Mongolia* or Montenegr* or Morocc* or Mozambi* or Myanma* or Burmese or Namibia* or Nepal* or Nicaragua* or Niger* or Nigeria* or Pakistan* or Palau* or Panama* or Papua New Guinea* or Paraguay* or Peru* or Philippines or Filipino or Romania* or Rwanda* or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leon* or Soloman Island* or Somalia* or South Africa* or Sudan* or Sri Lanka* or St Lucia* or St Vincent or Grenadines or Surinam* or Swazi* or Syria* or Tajikistan* or Tanzania* or Thai* or Timor* or Togo* or Tonga* or Tunisia* or Turk* or Turkmenistan* or Tuvalu* or Uganda* or Ukrain* or Uzbekistan* or Vanuatu* or Venezuela* or Vietnam* or West Bank or Gaza or Yemen* or Zambia* or Zimbabwe*).mp. | 3,750,176 |
| 13 | exp Africa/ or exp caribbean/ or exp caribbean islands/ or exp South America/ or exp Central America/ or exp Asia/ or exp Indian ocean/ or exp pacific ocean/ | 1,640,698 |

| | | |
|---|---|---|
| 14 | (africa* or asia* or caribbean or central america* or latin america* or south america* or melanesia* or micronesia* or polynesia*).mp. | 709,458 |
| 15 | (resource-limit* or resource-poor or low-resource* or limited-resource* or resource-constrain* or constrain*-resource* or under-resource* or poor*-resource* or resource-scarce* or scarce*-resource* or low-income or middle-income or lowincome or middleincome or (low adj3 middle-income)).mp. | 106,734 |
| 16 | ((developing or underdeveloped or under-developed or emerging or less-developed or least-developed or less-economically developed or least-economically developed or less-affluent or least-affluent or deprived or poor) adj (country or countries or nation or nations or region or regions or economy or economies)).mp. | 147,288 |
| 17 | ((developing or underdeveloped or under-developed or less-developed or least-developed) adj (population* or world)).mp. | 11,411 |
| 18 | (third-world* or thirdworld* or 3rd-world* or lmic or lmics or lami countr* or lalmi countr* or transitional countr*).mp. | 9,677 |
| 19 | (low* adj (gdp or gnp or gross domestic or gross national)).mp. | 346 |
| 20 | ((underserved or under served) adj (countr* or nation? or population*)).mp. | 4,543 |
| 21 | 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 | 4,538,802 |
| 22 | 10 and 21 | 2,828 |
| 23 | exp breeding/ | 50,299 |
| 24 | 22 not 23 | 2,824 |
| 25 | (exp animal/ or nonhuman/) not exp human/ | 7,135,403 |
| 26 | 24 not 25 | 2,700 |

### A.1.3 CINAHL search strategy

**CINAHL Complete**
Search date: 04 April 2020

Search strategy:

| # | Query | Results |
|---|---|---|
| S1 | (MH "Pregnancy+") | 221,733 |
| S2 | ( (pregnan* or maternal or gestation* or antenatal or prenatal or perinatal or natal or gravidit* or gravida* or multigravid* or primigravid*) ) OR ( (parity or parous or primiparity or multiparity or primiparous or multiparous) ) OR ( (expectant adj (mother* or wom#n or female*)) ) | 339,026 |
| S3 | S1 or S2 | 345,904 |
| S4 | (sleep* or slept or lying down or laid down or lie down or lay down or recumben* or reclining or reclined) | 78,377 |
| S5 | ((supine or prone) N3 (position* or posture*)) | 7,990 |
| S6 | ((lying or rest or rested or resting) and (position* or posture*)) | 4,216 |
| S7 | S4 or S5 or S6 | 88,178 |
| S8 | S3 and S7 | 4,146 |
| S9 | (MH "Developing Countries") | 19,300 |
| S10 | (Afghanistan* or Albania* or Algeria* or Angola* or Argentina* or Armenia* or Azerbaijan* or Bangladesh* or Belarus* or Beliz* or | 697,038 |

| | | |
|---|---|---|
| | Benin* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Botswan* or Brazil* or Bulgaria* or Burkina* or Burundi* or Cabo Verde* or Cape Verde* or Cambodia* or Cameroon* or Central African or Chad* or China or Chinese or Colombia* or Comor* or Congo* or Costa Rica* or Cote d'Ivoir* or Ivory coast or Cuba* or Djibouti* or Dominica* or Ecuador* or Egypt* or El Salvador* or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Georgia* or Ghana* or Grenad* or Guatemala* or Guinea* or Guyan* or Haiti* or Hondura* or Hungar* or India* or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kenya* or Kiribati* or Korea* or Kosov* or Kyrgyz Republic or Lao* or Leban* or Lesotho* or Liberia* or Libya* or Macedonia* or Madagascar* or Malawi* or Malaysia* or Maldiv* or Mali* or Marshall Island* or Mauritania* or Mauriti* or Mexic* or Micronesia* or Moldova* or Mongolia* or Montenegr* or Morocc* or Mozambi* or Myanma* or Burmese or Namibia* or Nepal* or Nicaragua* or Niger* or Nigeria* or Pakistan* or Palau* or Panama* or Papua New Guinea* or Paraguay* or Peru* or Philippines or Filipino or Romania* or Rwanda* or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leon* or Soloman Island* or Somalia* or South Africa* or Sudan* or Sri Lanka* or St Lucia* or St Vincent or Grenadines or Surinam* or Swazi* or Syria* or Tajikistan* or Tanzania* or Thai* or Timor* or Togo* or Tonga* or Tunisia* or Turk* or Turkmenistan* or Tuvalu* or Uganda* or Ukrain* or Uzbekistan* or Vanuatu* or Venezuela* or Vietnam* or West Bank or Gaza or Yemen* or Zambia* or Zimbabwe*) | |
| S11 | (MH ”Africa+”) or (MH ”Latin America”) or (MH “Central America+”) or (MH “South America+”) or (MH “West Indies+”) or (MH “Asia+”) or (MH “Atlantic Islands+”) or (MH “Indian Ocean Islands+”) or (MH “Melanesia+”) or (MH “Micronesia+”) or (MH “Samoa+”) | 462,910 |
| S12 | (africa* or asia* or caribbean or central america* or latin america* or south america* or melanesia* or micronesia* or polynesia*) | 149,335 |
| S13 | (resource-limit* or resource-poor or low-resource* or limited-resource* or resource-constrain* or constrain*-resource* or under-resource* or poor*-resource* or resource-scarce* or scarce*-resource* or low-income or middle-income or lowincome or middleincome or (low N3 middle-income)) | 42,073 |
| S14 | (developing or underdeveloped or under-developed or emerging or less-developed or least-developed or less-economically developed or least-economically developed or less-affluent or least-affluent or deprived or poor) N1 (country or countries or nation or nations or region or regions or economy or economies) | 32,103 |
| S15 | (developing or underdeveloped or under-developed or less-developed or least-developed) N1 (population* or world) | 2,823 |
| S16 | ( third-world* or thirdworld* or 3rd-world* or lmic or lmics or lami countr* or lalmi countr* or transitional countr* ) OR ( (low* N1 (gdp or gnp or gross domestic or gross national)) ) | 8,167 |
| S17 | ((underserved or under served) N1 (countr* or nation? or population*)) | 2,321 |
| S18 | S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 | 894,170 |
| S19 | S8 and S18 | 807 |

## A.1.4 Ovid Emcare search strategy

Database(s): **Ovid Emcare** 1995 to 2020 week 13
Search date: 04 April 2020

Search Strategy:

| # | Searches | Results |
|---|---|---|
| 1 | exp pregnancy/ | 117,279 |
| 2 | (pregnan* or maternal or gestation* or antenatal or prenatal or perinatal or natal or gravidit* or gravida* or multigravid* or primigravid*).mp. | 276,407 |
| 3 | (parity or parous or primiparity or multiparity or primiparous or multiparous).mp. | 14,959 |
| 4 | (expectant adj (mother* or wom#n or female*)).mp. | 626 |
| 5 | 1 or 2 or 3 or 4 | 281,390 |
| 6 | (sleep* or slept or lying down or laid down or lie down or lay down or recumben* or reclining or reclined).mp. | 77,544 |
| 7 | ((supine or prone) adj3 (position* or posture*)).mp. | 11,584 |
| 8 | ((lying or rest or rested or resting) and (position* or posture*)).mp. | 5,785 |
| 9 | 6 or 7 or 8 | 91,650 |
| 10 | 5 and 9 | 4,392 |
| 11 | developing countr*.mp. | 29,445 |
| 12 | (Afghanistan* or Albania* or Algeria* or Angola* or Argentina* or Armenia* or Azerbaijan* or Bangladesh* or Belarus* or Beliz* or Benin* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Botswan* or Brazil* or Bulgaria* or Burkina* or Burundi* or Cabo Verde* or Cape Verde* or Cambodia* or Cameroon* or Central African or Chad* or China or Chinese or Colombia* or Comor* or Congo* or Costa Rica* or Cote d'Ivoir* or Ivory coast or Cuba* or Djibouti* or Dominica* or Ecuador* or Egypt* or El Salvador* or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Georgia* or Ghana* or Grenad* or Guatemala* or Guinea* or Guyan* or Haiti* or Hondura* or Hungar* or India* or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kenya* or Kiribati* or Korea* or Kosov* or Kyrgyz Republic or Lao* or Leban* or Lesotho* or Liberia* or Libya* or Macedonia* or Madagascar* or Malawi* or Malaysia* or Maldiv* or Mali* or Marshall Island* or Mauritania* or Mauriti* or Mexic* or Micronesia* or Moldova* or Mongolia* or Montenegr* or Morocc* or Mozambi* or Myanma* or Burmese or Namibia* or Nepal* or Nicaragua* or Niger* or Nigeria* or Pakistan* or Palau* or Panama* or Papua New Guinea* or Paraguay* or Peru* or Philippines or Filipino or Romania* or Rwanda* or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leon* or Soloman Island* or Somalia* or South Africa* or Sudan* or Sri Lanka* or St Lucia* or St Vincent or Grenadines or Surinam* or Swazi* or Syria* or Tajikistan* or Tanzania* or Thai* or Timor* or Togo* or Tonga* or Tunisia* or Turk* or Turkmenistan* or Tuvalu* or Uganda* or Ukrain* or Uzbekistan* or Vanuatu* or Venezuela* or Vietnam* or West Bank or Gaza or Yemen* or Zambia* or Zimbabwe*) | 690,580 |
| 13 | (africa* or asia* or caribbean or central america* or latin america* or south america* or melanesia* or micronesia* or polynesia*).mp. | 151,429 |

| | | |
|---|---|---|
| 14 | (resource-limit* or resource-poor or low-resource* or limited-resource* or resource-constrain* or constrain*-resource* or under-resource* or poor*-resource* or resource-scarce* or scarce*-resource* or low-income or middle-income or lowincome or middleincome or (low adj3 middle-income)).mp. | 45,093 |
| 15 | ((developing or underdeveloped or under-developed or emerging or less-developed or least-developed or less-economically developed or least-economically developed or less-affluent or least-affluent or deprived or poor) adj (country or countries or nation or nations or region or regions or economy or economies)).mp. | 32,159 |
| 16 | ((developing or underdeveloped or under-developed or less-developed or least-developed) adj (population* or world)).mp. | 3,228 |
| 17 | (third-world* or thirdworld* or 3rd-world* or lmic or lmics or lami countr* or lalmi countr* or transitional countr*).mp. | 3,090 |
| 18 | (low* adj (gdp or gnp or gross domestic or gross national)).mp. | 72 |
| 19 | ((underserved or under served) adj (countr* or nation? or population*)).mp. | 1,931 |
| 20 | 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 | 811,037 |
| 21 | 10 and 20 | 778 |
| 22 | (exp animal/ or nonhuman/) not exp human/ | 442,079 |
| 23 | 21 not 22 | 775 |

### A.1.5 Global Medicus Index search strategy

Search date: 04 April 2020

Search Strategy:

| **Search** | **Results** |
|---|---|
| (“pregnancy” or “pregnant” or “maternal” or “gestation” or “parity” or “parturition” or “gravidity”)<br>AND<br>(“sleep” or “slept” or “sleeping” or “rest” or “lying down” or “laid down” or “sleeping position” or “sleeping posture”) | 154 |